COMPUTER SYSTEMS AND PLATFORM TECHNOLOGIES

IT0047

FEU ALABANG FEU DILIMAN FEU TECH

Technology Driven by Innovation

MODULE 1

Data Manipulation

MODULE 1. 3

Number System – Digital Codes, Parity, and Signed Number Representation

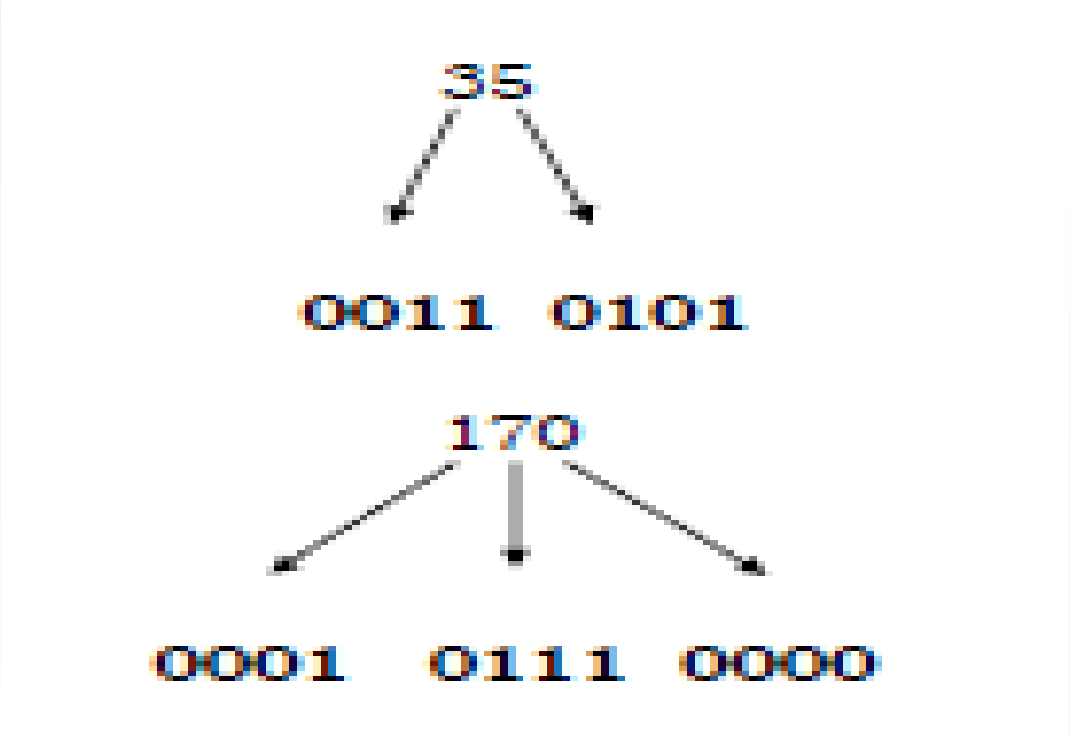
At the end of the lesson, the student should be able to :

- convert decimal numbers into digital codes (8421, 84-2-1, 2421, Excess-3, Biquinary, and Gray codes)
- identify decimal numbers represented by the digital codes
- distinguish Even from Odd parity
- Determine the different ways on signed binary representattion

The 8421 Code

- It is a type of **binary coded decimal (BCD) code. Binary coded decimal means that each decimal digit, 0 to 9, is represented by the binary code of four bits.**
- The designation 8421 indicates the binary weights of the four bits (23, 22, 21, 20).
- To express any decimal number in BCD, simply replace each decimal digit with the appropriate 4-bit code.

The 8421 Code



The 8421 Code

You should realize that, with four bits, sixteen numbers (0000 through 1111) can be represented but that, in the 8421 code, only ten of these are used. The six code combinations that are not used – 1010, 1011, 1100, 1101, 1110, and 1111 – are **invalid in 8421 code.**

The Excess-3 Code

This is unweighted code; its code assignment is obtained from the corresponding value of BCD after the addition of three.

The 84-2-1 and 2421 Codes

These are weighted code. The designation 8421 and 2421 indicates the binary weights of the four bits.

The Biquinary Code

- It is an example of a seven-bit code with error-detection properties. Each decimal digit consists of five 0's and two 1's placed in the corresponding weighted columns (5043210).
- During transmission of signals from one location to another, an error may occur. One or more bits may change value. A circuit in the receiving side can detect the presence of more (or less) than two 1's and if the received combination of bits does not agree with the allowed combination, an error is detected.

Digital Codes

Decimal	BCD	Excess 3	8 4 -2 -1	2 4 2 1	Biquinary
0	0000	0011	0000	0000	01 00001
1	0001	0100	0111	0001	01 00010
2	0010	0101	0110	0010	01 00100
3	0011	0110	0101	0011	01 01000
4	0100	0111	0100	0100	01 10000
5	0101	1000	1011	1011	10 00001
6	0110	1001	1010	1100	10 00010
7	0111	1010	1001	1101	10 00100
8	1000	1011	1000	1110	10 01000
9	1001	1100	1111	1111	10 10000

The Gray Code

- It is unweighted and is not arithmetic code: that is, there are no specific weights assigned to the bit positions.

- The important feature of the Fray Code is that *it exhibits only a single bit change from one code to the next.*

Binary-to-Gray Code Conversion

1. The most significant bit (left-most) in the Gray code is the same as the corresponding MSB in the binary number.
2. Going from left to right, add each adjacent pair of binary code bits to get the next Gray code bit. Discard carries.

Binary-to-Gray Code

Example:

1	+	→	0	+	→	1	+	→	1	+	→	0	Binary
↓			↓			↓			↓			↓	
1			1			1			0			1	Gray code

Decimal	Binary	Gray Code	Decimal	Binary	Gray Code
0	0000	0000	8	1000	1100
1	0001	0001	9	1001	1101
2	0010	0011	10	1010	1111
3	0011	0010	11	1011	1110
4	0100	0110	12	1100	1010
5	0101	0111	13	1101	1011
6	0110	0101	14	1110	1001
7	0111	0100	15	1111	1000

Gray Code-to-Binary Conversion

1. The most significant bit (left-most) in the Gray code is the same as the corresponding MSB in the binary number.
2. Add each binary code bit generated to the gray code bit in the next adjacent position. Discard carries.

Gray Code-to-Binary Conversion

Example:

1		1		0		1		0	Gray code
↓	+	↓	+	↓	+	↓	+	↓	
1		0		0		1		0	Binary

Parity

- It is one of the most common ways to achieve error detection.
- A *parity bit is an extra bit included with a message to make the total number of 1's transmitted either odd or even.*
- ***Odd Parity: The parity bit is chosen so that the total number of 1s is odd. Even Parity: The parity bit is chosen so that the total number of 1s is even.***
- It can detect odd number of errors but not even number of errors.

Parity

Examples:

	Odd Parity	Even Parity
0000	1	0
0001	0	1
0010	0	1
0011	1	0
0100	0	1
0101	1	0

7 ASCII bits + 1 Parity bit = 8 bits = 1 byte.

A p100 0001
(p = 0 for even parity.)| (p = 1 for odd parity.)

B p100 0010
(p = 0 for even parity.) (p = 1 for odd parity.)

C p100 0011
(p = 1 for even parity.) (p = 0 for odd parity.)

Parity

Error Detection Example

Send:
A (0100 0001) with even parity.

Receive:
Should receive (0100 0001) an even number of 1s.

If an odd number of 1s is received, there is an error.
0101 0001 - 1 error: Can be detected with parity.
0111 0001 - 2 errors: Cannot detect with parity.
0111 0011 - 3 errors: Can be detected with parity.
0111 0010 - 4 errors: Cannot detect with parity.
0101 1110 - 5 errors: Can be detected with parity.

Binary, Signed Integer Representations

Three systems are widely used for representing both positive and negative numbers:

- 1. Sign and Magnitude
- 2. 1's – complement
- 3. 2's – complement

- For all three systems, the leftmost bit is 0 for positive numbers and 1 is for negative numbers

Binary, Signed Integer Representations

$b_3\ b_2\ b_1\ b_0$	Sign and Magnitude	1's Complement	2's Complement
0000	+ 0	+ 0	+ 0
0001	+ 1	+ 1	+ 1
0010	+ 2	+ 2	+ 2
0011	+ 3	+ 3	+ 3
0100	+ 4	+ 4	+ 4
0101	+ 5	+ 5	+ 5
0110	+ 6	+ 6	+ 6
0111	+ 7	+ 7	+ 7
1000	- 0	- 7	- 8
1001	- 1	- 6	- 7
1010	- 2	- 5	- 6
1011	- 3	- 4	- 5
1100	- 4	- 3	- 4
1101	- 5	- 2	- 3
1110	- 6	- 1	- 2
1111	- 7	- 0	- 1

Be reminded that:

- Positive values have identical representations in all systems.
- In the sign-and-magnitude system, negative values are represented by changing the MSB of the positive representation to 1.

Be reminded that:

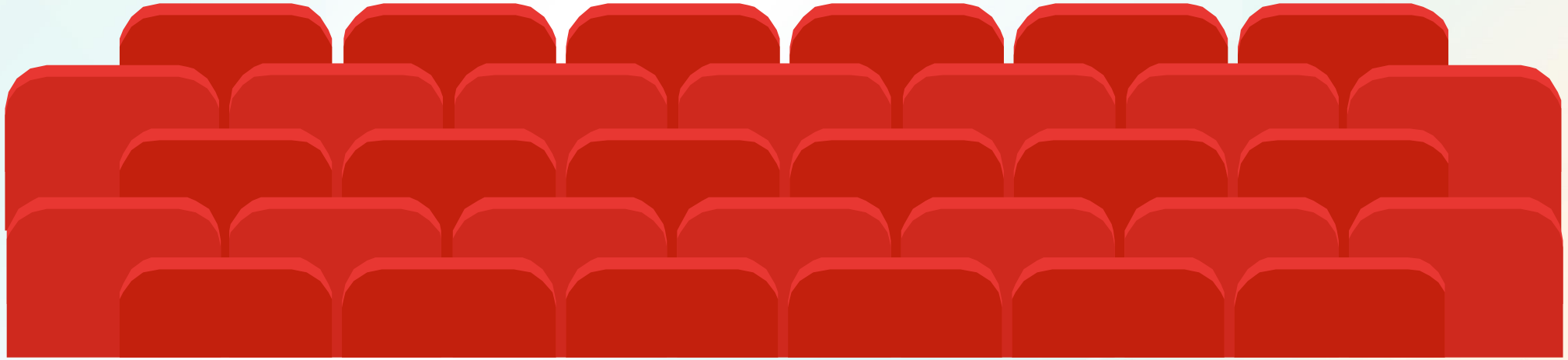
- In the 1's-complement system, negative values are obtained by complementing each bit of the corresponding positive representation.

- In the 2's-complement system, a negative number is obtained by subtracting the positive number from 2n. Hence, the 2's-complement representation is obtained by adding 1 to the 1's-complement representation.

Overflow and Underflow

- Overflow (Addition)
  - MSB - Carry, No Carry
  - MSB – No Carry, Carry
- Underflow (Subtraction)
  - MSB – Borrow, No Borrow
  - MSB – No Borrow, Borrow

Q&A SESSION

ASK ANY QUESTION RELATED TO OUR TOPIC FOR TODAY.

THANK YOU